

Sidra Ghulam Hussain

Software Engineer

Personal Information

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Professional Summary

BS Software Engineering student and developer with hands-on experience building production-style MERN stack and real-time web applications, including e-commerce platforms, social media apps, and WebRTC-based video conferencing tools. Certified completion of a Full Stack Development internship at CodeAlpha (4 independent projects covering authentication, database design, RESTful APIs, and real-time architecture with Socket.IO and WebRTC), and currently undertaking a second CodeAlpha internship in Frontend Development. Strong foundation in C++, Python, and object-oriented programming, with proven ability to deliver working end-to-end applications, not just static prototypes.

Technical Skills

- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend:** Node.js, Express.js, RESTful API Development
- **Databases:** MongoDB, SQLite
- **Real-Time & Networking:** Socket.IO, WebRTC, WebSockets
- **Authentication & Security:** JWT, Password Hashing, AES-256-GCM Encryption
- **Programming Languages:** C++, Python, C, OOP, Shell Scripting
- **Tools & Platforms:** Git, GitHub, Figma, Cisco Packet Tracer, MS Office, Ubuntu Terminal, Canva, Adobe Photoshop
- **Soft Skills:** Teamwork, Communication, Project Management, Problem-Solving

Work Experience / Internship

Frontend Development Intern – CodeAlpha (Remote), Jul 2026 – Aug 2026

- Successfully completed the Frontend Development Internship at CodeAlpha and received a Certificate of Completion (Student ID: CA/DF1/222069).
- Demonstrated dedication, collaboration, and the ability to quickly adapt to new technologies during the internship period.
- Received a Letter of Recommendation from CodeAlpha recognizing strong analytical and teamwork skills.

Full Stack Development Intern – CodeAlpha (Remote), Jun 2026 – Jul 2026

- Independently designed, developed, and deployed 4 full-stack web applications end-to-end — spanning e-commerce, social media, real-time project management, and video conferencing — exceeding the internship's required 2-3 project minimum.
- Implemented secure authentication (JWT, hashed passwords) and data protection (AES-256-GCM client-side encryption) across multiple production-style applications.
- Built real-time, multi-user features using Socket.IO and WebRTC, including live Kanban sync, peer-to-peer video calls, and instant notifications, validated through live multi-user testing sessions.
- Successfully completed the internship (Certificate ID: CA/DF1/162482) and received a Letter of Recommendation from CodeAlpha citing strong analytical skills and collaboration.

Intern, Web Development – Arch Technologies, 2026 – Present

- Assisted in developing multiple client websites using HTML, CSS, and JavaScript, including a music player app, a task management app, and a personal portfolio site (see Projects section).

Intern, Graphic and Visual Design – GAO Tek Inc., 2025

- Designed logos and marketing visuals for two websites using Canva; collaborated with senior developers on visual design integration and image editing.

Key Projects – Full Stack Development (CodeAlpha)

Verve Market – Full-Stack E-Commerce Platform

Tech Stack: React, Node.js, Express.js, MongoDB, JWT

- Developed a full-stack e-commerce store with product search, filtering, sorting, and a persistent shopping cart that survives page refresh.
- Engineered live order-processing logic with real-time inventory validation and stock updates, replacing a static "Buy Now" button with functional checkout and order-tracking codes.
- Built a custom boutique-style UI from scratch without any UI framework; secured user accounts with JWT authentication and hashed passwords.

GitHub: [CodeAlpha VerveMarket](#) | Demo: [Demo Video](#)

Pulse Circle – Social Media Web Application

Tech Stack: HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, JWT

- Built a social platform with editable user profiles, posts, likes, and threaded comments.
- Designed a real follow system where the feed prioritizes followed users' content, with automatic fallback to a public feed for new users.
- Implemented instant, no-reload updates for follow/unfollow, like, and comment actions using client-side state management.

GitHub: [CodeAlpha PulseCircle](#) | Demo: [Demo Video](#)

Flowdeck – Real-Time Project Management Tool

Tech Stack: Node.js, Express.js, Socket.IO, SQLite, JWT

- Developed a Trello-style Kanban board for teams to create projects, invite members, and manage tasks with drag-and-drop cards synced instantly across all connected clients.
- Implemented task assignment, priorities, due dates, threaded comments, live user-presence indicators, and real-time push notifications via Socket.IO (WebSockets).
- Validated real-time synchronization across concurrent multi-user sessions.

GitHub: [CodeAlpha Flowdeck](#) | Demo: [Demo Video](#)

Confero – Real-Time Video Conferencing & Collaboration App

Tech Stack: WebRTC, Node.js, Socket.IO, JWT, AES-256-GCM

- Built a multi-user video conferencing platform on a peer-to-peer WebRTC mesh with live screen sharing and a real-time collaborative whiteboard.
- Implemented serverless peer-to-peer file transfer via WebRTC DataChannels and client-side AES-256-GCM message encryption, so the server only ever relays ciphertext.
- Added JWT authentication with optional password-protected meeting rooms.

GitHub: [CodeAlpha Confero](#) | Demo: [Demo Video](#)

Frontend Development Projects (Arch Technologies)

AuroraWave – Music Player Web App

Tech Stack: HTML, CSS, JavaScript

- Built a browser-based music player with a song library, album artwork display, and custom playback controls (play, pause, seek) using vanilla JavaScript DOM manipulation.

GitHub: [AuroraWave-Music-App](#) | Demo: [Repository](#)

Todo App – Task Management Web App

Tech Stack: HTML, CSS, JavaScript

- Developed a task management application allowing users to add, complete, and delete tasks, with dynamic DOM updates and client-side state handling.

GitHub: [sidra-todo-app](#) | Demo: [Repository](#)

Personal Portfolio – Responsive Portfolio Website

Tech Stack: HTML, CSS

- Designed and built a responsive personal portfolio website to showcase projects, technical skills, and resume.

GitHub: [sidra-portfolio](#) | Demo: [Repository](#)

Other Academic Projects

- Digital Calculator Design – Logic circuits and embedded coding with DLD kit.
- Room Vista (AR-Based Mobile App) – Designed prototype with Figma, documented SRS and SDD.

- Banking System (C++) – Console-based system with secure login and operations.
- ATM & Branch Integration (C++) – OOP-based simulation with modular design.
- Infix to Postfix Converter (C++) – Custom stack implementation for expression evaluation.
- University Networking System – Configured routers, switches, and servers using Cisco Packet Tracer.
- Making a Story Book (Canva) – Designed images for a story book using AI tools like Canva and Microsoft Copilot; created a complete story book including quiz, puzzle, and glossary.
- Stack Operation Visualizer in 8086 Architecture (Python, COAL) – Built a visualizer to demonstrate stack operations at the assembly level.
- 3D Car Racing Game Prototype (Figma) – Designed a prototype showing car selection, level selection, settings, and results.
- 3D Car Racing Game (C++ Console-Based) – Coded car selection, level selection, settings, and results in a console window.

Education

- BS Software Engineering – Air University, Multan Campus (2023–Present, CGPA: 3.72/4.0)
- Intermediate (Pre-Engineering) – Pakistan College Multan (2020–2022, Marks: 947/1100)
- Matriculation (Science Group) – Al-Rehmat Higher Secondary School Multan (2018–2020, Marks: 1045/1100)

Certifications / Achievements

- Certificate of Completion, Full Stack Development Internship – CodeAlpha (Student ID: CA/DF1/162482, Jun-Jul 2026)
- Letter of Recommendation, Full Stack Development Internship – CodeAlpha (Student ID: CA/DF1/162482, Issued 21 Jul 2026)
- Internship Offer Letter, Full Stack Development – CodeAlpha (Jun-Jul 2026)
- Certificate of Completion, Frontend Development Internship – CodeAlpha (Student ID: CA/DF1/222069, Aug 2026)
- Letter of Recommendation, Frontend Development Internship – CodeAlpha (Student ID: CA/DF1/222069, Issued 1 Sep 2026)
- Internship Offer Letter, Frontend Development – CodeAlpha (Jul-Aug 2026)
- Internship Offer Letter, Web Development – The Sky Gen Internship Program 2026 (Sep-Nov 2026)
- Certificate of Completion, Graphic and Visual Design Internship – GAO Tek Inc. (2025)
- Certificate of Training, Graphic Design – GAO Tek Inc. (2025)
- Certificate of Training, Digital Marketing – GAO Tek Inc. (2025)
- Internship Offer Letter, Web Development – Arch Technologies (2025)
- AI for Beginners Online Course – HP Life (2025)
- Resume Writing and Job Interviewing Online Course – HP Life (2025)
- Content Writing Course – Air University Multan Campus (2024)
- Extraordinary Performance in Matriculation – Legend College Multan (2021)

Languages

- Urdu – Native/Fluent
- Punjabi – Conversational
- English – Working Proficiency

References

Available upon request.